Prescripto AI 2.0 — Complete Technical Stack Inventory

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Canonical Spec Reference: backend/requirements.txt, web/package.json, docker-compose.yml
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1. Backend Core & Application Framework

Component	Library / Binary	Version in Repo	Exact Purpose	File References
Language Runtime	Python	`3.11` / `3.12`	Core backend execution runtime	Dockerfile, .github/workflows/ci.yml
API Framework	FastAPI	`0.115.0`	Asynchronous REST API routing, request validation, OpenAPI auto-generation	backend/prescripto/api/main.py
ASGI Engine	Starlette / Uvicorn	Uvicorn `0.32.0`, Starlette `0.38.6`	HTTP server execution, reverse-proxy header propagation (`--proxy-headers`)	backend/prescripto/api/main.py
Data Validation	Pydantic v2	`>=2.10.0,<3.0.0`	Strict schema serialization, input validation, custom field validators	backend/prescripto/api/v1/schemas/
Settings Management	Pydantic Settings	`2.6.0`	Multi-source environment loading (`.env` , `backend/.env` , system env vars)	backend/prescripto/config/settings.py
Structured Logging	Structlog	`24.4.0`	High-performance JSON logging with custom Zero-PHI whitelist processors	backend/prescripto/audit/logger.py
HTTP Client	HTTPX	`0.28.1`	Asynchronous HTTP requests, TestClient execution, integration test harness	backend/requirements.txt

2. Relational Database & Persistence Layer

Component	Library / Binary	Version in Repo	Exact Purpose	File References
Database Engine	PostgreSQL	`16-alpine`	Relational storage, ACID transactions, `FOR UPDATE SKIP LOCKED` worker queue	docker-compose.yml, render.yaml
Lightweight Engine	SQLite3	Native Python 3.11	Hermetic unit test execution and isolated developer environment	backend/prescripto/db/session.py
ORM Framework	SQLAlchemy	`2.0.35`	Object-Relational Mapping, connection pooling, database session management	backend/prescripto/db/session.py
Schema Migrations	Alembic	`1.20.0`	Versioned schema migration scripts (`0001` , `0002` , `0003`)	backend/prescripto/db/migrations/
Postgres Driver	Psycopg2-binary	`2.9.10`	Low-level C-based PostgreSQL database adapter	backend/requirements.txt

3. Machine Learning, Computer Vision & Perception

Component	Library / SDK	Version in Repo	Exact Purpose	File References
Primary Vision SDK	Google GenAI SDK	`2.14.0`	Cloud multimodal vision extraction using `gemini-3.5-flash` and `gemini-3.1-flash-lite`	backend/prescripto/ml/recognition_providers.py
Local OCR Runtime	RapidOCR (ONNX)	`rapidocr-onnxruntime`	Offline CPU-based handwriting/printed OCR fallback when Gemini is unavailable	backend/prescripto/ml/rapid_runtime.py
Image Processing	Pillow (PIL)	`10.4.0`	Image decoding, resizing, thumbnail generation, color-space conversion	backend/prescripto/pipeline/engine.py
Computer Vision	OpenCV Headless	`4.10.0.84`	Laplacian blur variance calculation, contrast and exposure threshold gates	ml/ocr/quality_check.py
Detection Backbone	PaddleOCR DBNet++	Configured in `ml/`	Bounding box text detection candidate fine-tuned on prescription margins	ml/ocr/PP-OCRv6_small_rxhandbd.yml

4. Security, Cryptography & Object Storage

Component	Library / Protocol	Version in Repo	Exact Purpose	File References
JWT Tokens	PyJWT	`2.9.0`	Asymmetric RS256 token encoding and decoding with expiration checks	backend/prescripto/auth/crypto.py
Cryptographic Lib	Cryptography	`43.0.3`	2048-bit RSA keypair generation, PKCS8 serialization, and verification	backend/prescripto/config/settings.py
Password Hashing	Bcrypt	`4.2.0`	Salted SHA-256 bcrypt hashing for clinician and administrator credentials	backend/prescripto/auth/crypto.py
Object Store SDK	Boto3 / Botocore	`1.35.40`	S3-compatible object storage communication, bucket lifecycle, presigned URLs	backend/prescripto/storage/client.py
Signed URL Auth	HMAC-SHA256	Native Python `hmac`	Ephemeral 60-second presigned URLs for local storage streaming	backend/prescripto/storage/client.py

5. Frontend UI & Client Technology

Component	Technology	Version in Repo	Exact Purpose	File References
UI Framework	React	`^19.3.0`	Declarative component hierarchy, virtual DOM, reactivity	web/package.json
Language	TypeScript	`~6.0.2`	Compile-time type safety, interface contracts matching backend DTOs	web/tsconfig.json
Build & Bundler	Vite	`^8.3.0`	Instant HMR development server, optimized rollup asset chunking	web/vite.config.ts
Styling Engine	Tailwind CSS v4	`^4.3.3`	Utility-first responsive design, humanized clinical light color tokens	web/src/index.css
Client Router	React Router DOM	`^7.18.4`	Single-page application routing, route guards (`RequireRole`), SPA fallback	web/src/App.tsx
HTTP Interceptor	Axios	`^1.20.0`	Global token injection, auto-refresh queue on 401, UUID `X-Request-ID` generation	web/src/api/client.ts
Iconography	Lucide React	`^1.52.0`	Medical iconography (pills, stethoscopes, warnings, shields, arrows)	web/src/components/
Voice Synthesis	Web Speech API	Native Browser	High-clarity spoken audio readouts for patients in 4 Indian languages	web/src/pages/PatientPortalPage.tsx

6. Infrastructure, Containerization & CI/CD

Platform / Tool	Role in Project	Configuration File	Verified Behavior
Docker Multi-Stage	Compiles React Vite into static dist and bundles with Python runtime	Dockerfile	Validated on Debian 12 slim
Docker Compose	Local orchestration: `api` , `worker` , `postgres` , `minio` , `createbuckets`	docker-compose.yml	Verified with ports 8000, 5435, 9000, 9001
Render Cloud	Backend API hosting, managed PostgreSQL, auto-deploy from `main`	render.yaml	Verified at `prescripto-2-0.onrender.com`
Vercel	Frontend static site deployment with SPA routing rewrites	vercel.json, web/vercel.json	Verified client-side URL routing
GitHub Actions	Automated CI pipeline: dependency installation, linting, and pytest suites	.github/workflows/ci.yml	Runs on push to `main`